

Software Engineering

(SE)



Planning



Analyze



Design



Development

Software Engineering



Implementation



Testing



Validation and
Verification



Maintenance

- Software is a program or set of programs containing instructions that provide the desired functionality. Engineering is the process of designing and building something that serves a purpose efficiently and cost-effectively

- It is the systematic process of designing, developing, testing, and maintaining software using techniques like requirements analysis, design, testing, and maintenance.
- It ensures software is high-quality, reliable, and maintainable, especially for large software systems.
- Software Engineering improves efficiency, quality, and budget management, helping deliver software on time, within budget, and meeting all requirements.
- It continuously adopts new tools and methodologies to handle complexity and evolving technologies.

Key Principles

- Modularity: Divide software into small, independent parts that are easy to develop and test.
- Abstraction: Show only what is necessary and hide internal implementation details.
- Encapsulation: Keep data and related functions together and protect them from outside access.
- Maintenance: Regularly update software to fix bugs, improve features, and enhance security.
- Testing: Check the software to ensure it works correctly and meets requirements.
- Design Patterns: Use proven solutions for common software design problems.
- Agile Methodologies: Develop software in small steps with frequent feedback and flexibility.
- Continuous Integration & Deployment (CI/CD): Frequently merge code changes and automatically deploy updates.

Objectives or Attributes

- **Maintainability:** It should be feasible for the software to evolve to meet changing requirements.
- **Efficiency:** The software should not make wasteful use of computing devices such as memory, processor cycles, etc.
- **Correctness:** A software product is correct if the different requirements specified in the SRS Document have been correctly implemented.
- **Reusability:** A software product has good reusability if the different modules of the product can easily be reused to develop new products.
- **Testability:** Here software facilitates both the establishment of test criteria and the evaluation of the software concerning those criteria.

***SRS -- Software Requirement Specification**

- **Reliability:** It is an attribute of software quality. The extent to which a program can be expected to perform its desired function, over an arbitrary time period.
- **Portability:** In this case, the software can be transferred from one computer system or environment to another.
- **Adaptability:** In this case, the software allows differing system constraints and the user needs to be satisfied by making changes to the software.
- **Interoperability:** Capability of 2 or more functional units to process data cooperatively.

What Tasks do Software Engineers do?

- **Requirement Analysis:** Collaborating with stakeholders to understand and gather the requirements to design and develop software solutions.
- **Design and Development:** Creating well-structured, maintainable code that meets the functional requirements and adheres to software design principles.
- **Testing and Debugging:** Writing and conducting unit tests, integration tests, and debugging code to ensure software is reliable and bug-free.

- **Code Review:** Participating in code reviews to improve code quality, ensure adherence to standards, and facilitate knowledge sharing among team members.
- **Maintenance:** Updating and maintaining existing software systems, fixing bugs, and improving performance or adding new features.
- **Documentation:** Writing clear documentation, including code comments, API documentation, and design documents to help other engineers and future developers understand the system.

SE Type

- PC Based
 - Vision Image Processing (ADE)
 - Vision Application UI
 - Handler (Machine motion control)
- PLC
 - Handler (Machine motion control)
 - HMI (Human-Machine Interface)

ADE (Algorithm Design Engineer)

- **MVTec HALCON:** A comprehensive, global standard in industrial machine vision. It is known for state-of-the-art 3D vision, deep learning tools, and broad multi-core/GPU acceleration.
- **Cognex VisionPro:** A leading PC-based vision software suite for industrial applications. It flawlessly combines robust traditional tools (like geometric object location and caliper measurements) with deep learning.
- **Zebra Aurora Vision:** A highly visual, studio-based software designed by machine vision experts. It allows enterprises to easily integrate AI-based algorithms for complex surface defect detection and character reading.

- **Open eVision (by Euresys):** A hardware-independent image processing and analysis library compatible with any GigE Vision or USB3 Vision camera. It supports 3D, deep learning, and accurate sub-pixel measurements.
- **NI Vision Development Module:** A robust set of vision libraries designed for seamless integration with National Instruments hardware (like LabVIEW) and industrial controllers for rapid inspection system deployment.
- **OpenCV:** An open-source computer vision and machine learning software library widely used for real-time applications.

Image Processing

- **Edge detection/extraction** – In an image, an edge marks a distinct change in intensity. Such changes mark discontinuities in depth, orientation, and material, and allow for finding the boundaries of objects.
- **Pixel counting** – A straightforward algorithm can count the number of pixels at each grayscale level in a region of interest or the overall image, providing information used by other functions.
- **Blob detection** – Different regions can be identified, providing information used by other functions.

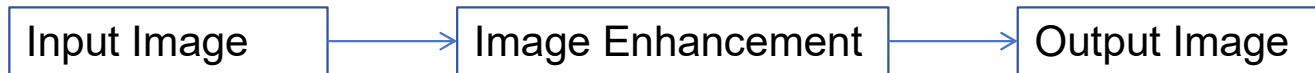
- **Pattern/template matching** – Guiding and positioning functions use pattern matching or template matching on parts and products to determine object location and orientation.
- **Color analysis** – Where object color is an important feature, there are several color tools, including color extraction and segmentation, and color matching.

Image Processing



Image Processing	Methods
Image Enhancement	Point Operation, Mask Operation, Histogram Operation
Image Segmentation	Threshold, Classification, Edges, Regions
Image Transformation	Translation, Rotation, Scaling, Shearing, Reflect, Invert, Affine
Image Correction	Distortion, Flat Field, Color, Blur(Defocus/Motion) Correction
Image Interpolation	Nearest Neighbor, Bilinear, Constant, Weighted, Bicubic
Image Compression	Lossless, Lossy

Image Enhancement



Operation	Techniques of Image Filtering	Example of Commonly Used Filter/ Methods
Point Operation	Arithmetic Operation	Brightness Enhancement, Invert Image, Contrast Stretching, Gray Level Slicing, Binary Threshold
Mask Operation	Noise Reduction	Mean, Median, Gaussian /Bilateral, Wiener
Mask Operation	Edge detection	Laplacian, Fast Fourier Transform(Freq), Canny, Sobel, Kirsch, Roberts
Mask Operation	Smoothing	Mean Filter, Median Filter, Gaussian Filter, Wiener, Low Pass Filter
Mask Operation	Sharpening	High Pass Filter
Mask Operation	Feature Enhancement	Morphology (Erode, Dilate, Open, Close)
Histogram Operation	Contrast Improvement	Histogram Equalization



Image Segmentation



Techniques of Image Segmentation	Example of Commonly Used Methods
Threshold	Binary Threshold, Dual Threshold, Adaptive Threshold, Dynamic Threshold
Classification	Classification + Localization, Object Detection, Semantic Segmentation, Instance Segmentation
Edges	Prewitt, Sobel, Robert, Kirsch, Derivative Gauge
Region	Region Splitting, Region Growing



Image Transformation

- **Translation:** Shifting an image from one location to another along the x and y axes.
- **Scaling:** Resizing an image (zooming in or out) by expanding or contracting the pixel grid.
- **Rotation:** Turning an image around a fixed central point by a specific angle.
- **Reflection:** Flipping the image horizontally (along the x-axis) or vertically (along the y-axis).
- **Shearing:** Shifting pixels either horizontally or vertically to create a slanted or skewed effect.
- **Perspective & Affine Transforms:** Warping an image to simulate a different viewpoint (e.g., a bird's-eye view) or correcting camera lens distortion



(a) Input



(b) Translate



(c) Scale



(d) Rotate



(e) Shear



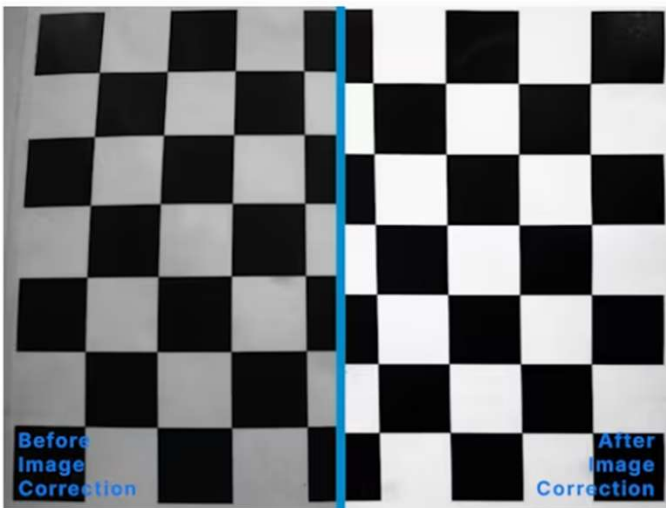
(f) Affine

Image Correction

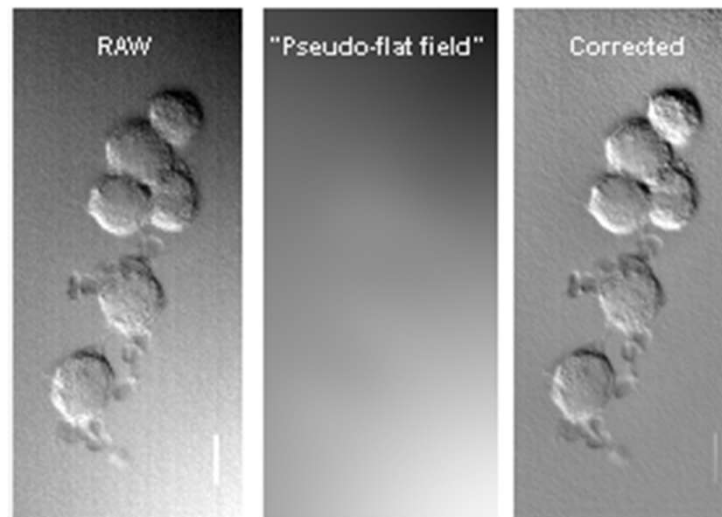
- Distortion Correction
- Flat Field Correction
- Color Correction
- Blur Correction (Defocus/Motion)



Blur Correction



Distortion Correction



Flat Field Correction



Color Correction

Image Interpolation

Interpolation method	Quality	Runtime (relative to <i>nearest_neighbor</i>)	Recommended usage
<i>nearest_neighbor</i>	low	1	Only for time-critical applications
<i>bilinear</i>	high (for upscaling)	3	If downscaling must not be slower than upscaling
<i>constant</i>	enhanced for downscaling	3 (upsampling) 10 (downscaling)	If the image quality should be consistent for downsampled and upsampled images
<i>weighted</i>	best for downscaling	3 (upsampling) 20 (downscaling)	If the image quality of downsampled images is critical for the application
<i>bicubic</i>	very high (for upscaling)	6	If the image quality of upsampled images is critical for the application

Table 2.1: Interpolation methods for image transformations and their characteristics with respect to quality, runtime, and recommended usage. Note that the runtime factors may vary depending on the downscaling factor and on your hardware.

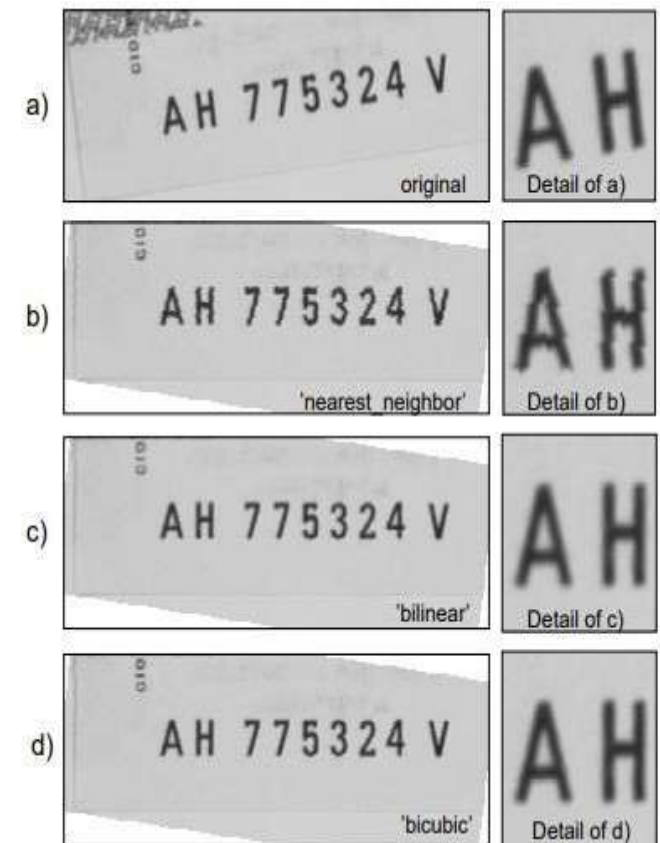
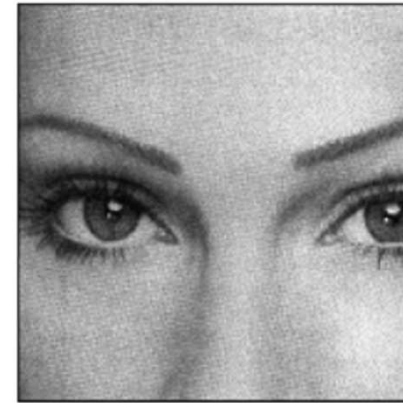


Image Compression

- Lossless - Reduces file size without discarding any information. The decompressed image is an exact replica of the original. (examples: PNG, TIFF)
- Lossy - Permanently removes imperceptible or redundant data to achieve drastically smaller file sizes. (examples: JPEG)



(a) Original Image



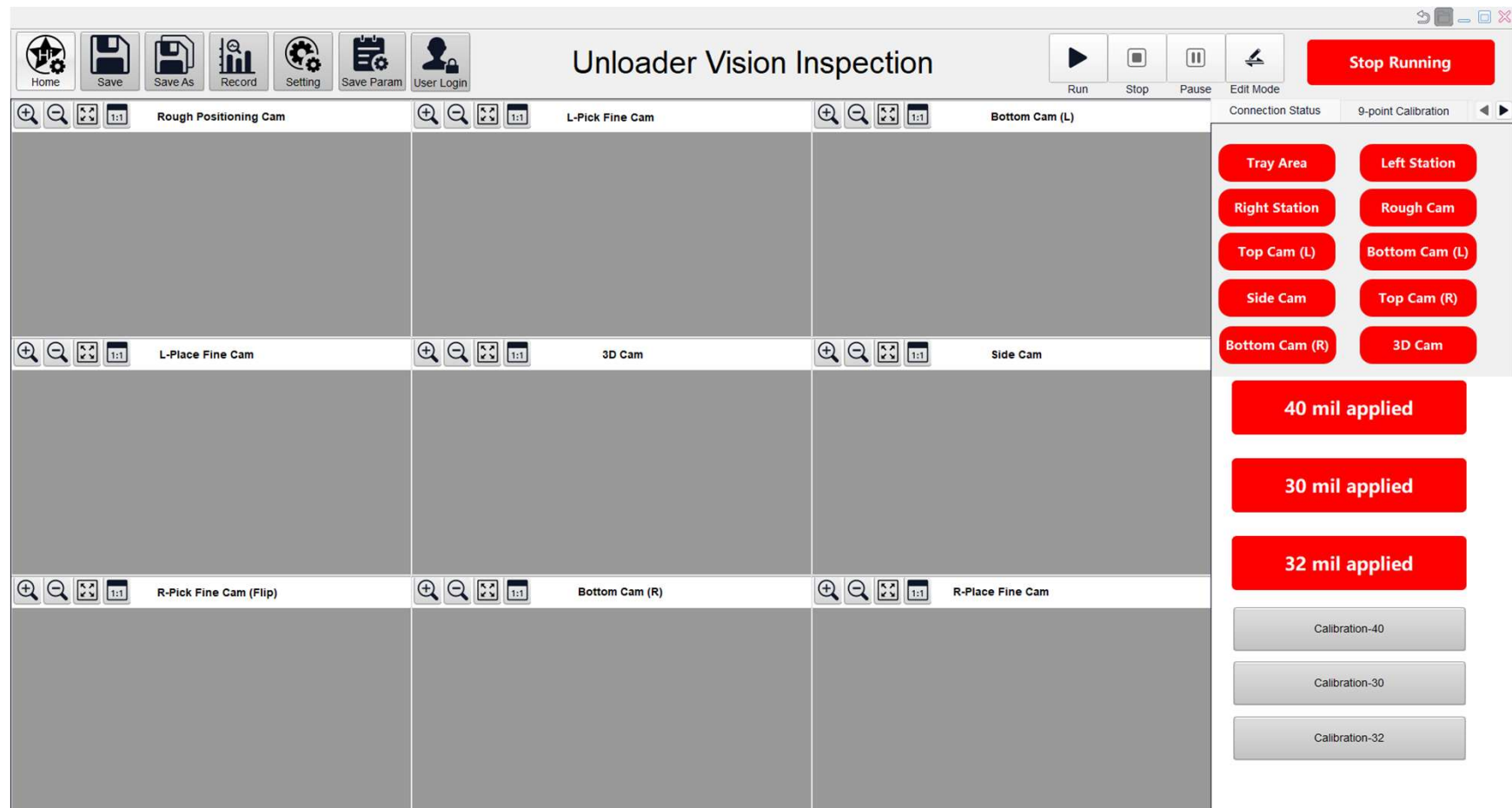
(b) Image with 10:1 compression



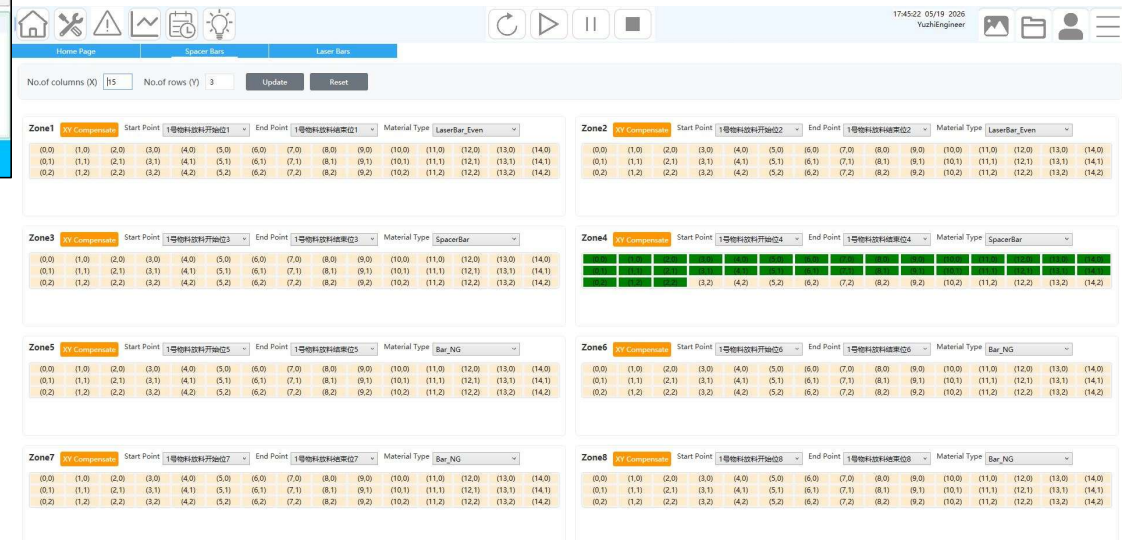
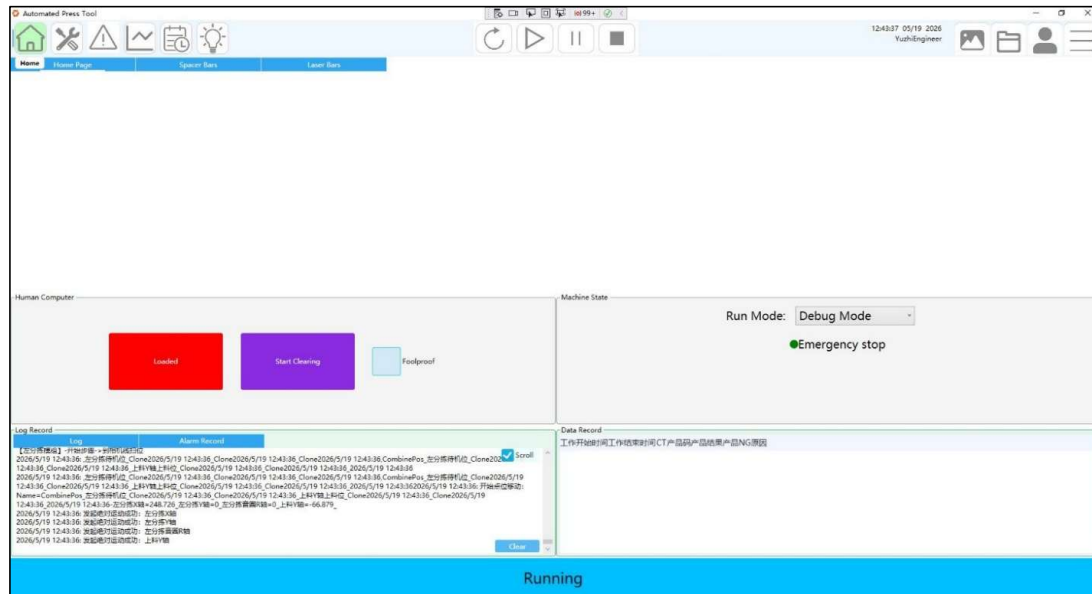
(c) Image with 45:1 compression

Example of a JPEG distortion .Figure (a) shows the original image, while (b) and (c) shows the image using compression ratios of 10:1 and 45:1.

Vision Application UI



Handler (Machine motion control)



PLC (Programmable Logic Controller)

- Process of creating instructional code that tells industrial automation machines and systems how to operate.
- The program reads signals from physical sensors (inputs) and uses logic to control motors, valves, and actuators (outputs).

How PLC works

- PLCs operate in a continuous cycle called a "scan".
 - **Input Scan:** Reads the state of all connected input devices.
 - **Program Execution:** Executes the programmed instructions to determine the target state of outputs.
 - **Output Scan:** Updates the output signals based on the logic
 - **.Housekeeping:** Performs safety checks and diagnostics, then restarts the cycle.

Common Programming Languages

- **Ladder Diagram (LD):** A graphical language that mimics traditional relay circuit schematics. It is the most popular language because it is highly visual and easy to troubleshoot.
- **Function Block Diagram (FBD):** A graphical language that uses "black box" logic blocks. It is ideal for safety logic and data reuse.
- **Structured Text (ST):** A high-level, text-based language similar to C or Python. It is best for complex math, data manipulation, and looping.

- **Sequential Function Chart (SFC):** A flowchart-style graphical language used to break complex, multi-state processes into manageable sequences.
- **Instruction List (IL):** A low-level, assembler-style textual language. It is largely deprecated in modern systems

PLC brand

- **Siemens:** Leading the market globally, Siemens is highly regarded for its scalability and integration. Its Simatic S7 series (e.g., S7-1200 and S7-1500) is widely used in smart factories, automotive sectors, and heavy industry.
- **Allen-Bradley (Rockwell Automation):** A dominant force in North America and popular globally for its real-time control and easy setup. Their ControlLogix system is renowned for factory and process automation.
- **Mitsubishi Electric:** Highly prominent in Asia, particularly in SMEs, packaging, and Food & Beverage. Known for the MELSEC series, this brand offers cost-effective, modular, and upgradeable solutions.

- **Omron:** A strong player in the semiconductor and precision electronics industries. Their Sysmac series is compact and highly regarded for robotics and motion control.
- **Schneider Electric:** Known for energy efficiency and IoT-ready solutions. Their Modicon M580 and M262 series are frequently used in hyperscale data centers and material handling.
- Other brand like Beckhoff, Keyence, Inovance

HMI

- Involves designing graphical interfaces that allow operators to monitor, control, and interact with industrial machines and systems.
- It translates a PLC's (Programmable Logic Controller) raw data into visual elements like buttons, charts, and alarm logs.

Key Concepts

- **Interface vs. Control:** HMIs do not run the actual machine control logic; that job belongs to the PLC. The HMI simply displays system parameters, allows user inputs, and writes those commands to the PLC.
- **Screen Design:** Designing intuitive navigation, clear visual indicators, and easily readable data displays is essential for operator safety and efficiency.
- **Linking Variables:** Every object on the HMI screen (like a lamp or push-button) must be linked to a specific memory address or tag in the PLC program to function correctly.

Popular Software Platforms

- **Weintek / Maple Systems:** Uses EasyBuilder Pro (EBPro), a widely used, highly accessible software with robust simulation tools.
- **Siemens:** Configured using TIA Portal (WinCC).
- **Allen-Bradley:** Utilizes FactoryTalk View.
- **AutomationDirect:** Configured using C-more programming software.
- **Open Platforms:** Ignition is widely used for PC-based industrial applications and offers free trials and training.